SYNAPTICSTEEL / LAB RESEARCH

Conceptual Compression and Context Fidelity in Human–LLM Collaboration

A Research Framework for Assumption Propagation and Conversational Drift

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Classification: Theoretical research framework and proposed experiment. Not peer-reviewed. No original empirical results.

ABSTRACT

Humans often communicate complex ideas through a few conceptual landmarks while omitting relationships that are familiar to them but unavailable to a listener. Large language models (LLMs) can reconstruct missing relationships, but may also introduce plausible, unsupported inferences. This paper proposes that unapproved conversational drift may depend less on compressed communication itself than on unverified model inferences being promoted into accepted context during subsequent turns. It distinguishes biological conceptual compression from LLM reconstruction, recoverable omissions from genuinely unavailable information, and authorized goal changes from unapproved drift. An intent-graph representation and proposed measures of task-weighted distortion and context debt are introduced, together with five falsifiable hypotheses and a controlled evaluation design. The causal mechanism remains a research hypothesis, not a demonstrated finding.

Keywords: conceptual compression; large language models; human–AI collaboration; context fidelity; assumption propagation; inferential distance; context debt; industrial intelligence.

1. Research question and claim boundary

A person may express a complex idea as A → C → J → X → Z, leaving many intermediate relationships implicit. Those links may be supported by learned schemas and shared knowledge, or may be unresolved. This sequence is an illustration of communication, not a claim that the brain secretly computes a complete linear chain from A through Z. Complex knowledge is more naturally represented as a network of concepts, dependencies, constraints, and goals.

A language model faces a different challenge: it receives the utterance, available documents or stored context, and learned linguistic representations, but it cannot directly inspect the speaker's private experiences. Its reconstruction may be correct, partially supported, or unjustified. The risk

increases when later turns silently promote an unjustified inference into a confirmed premise.

> ***Central hypothesis: In multi-turn human–LLM tasks, consequential, unsupported inferences that are promoted into accepted context will predict greater unapproved task drift than the amount of compressed language alone.***

Four distinct events must be separated: omission, model reconstruction, assumption promotion, and deviation from the currently approved task. The present paper proposes their relationship; it does not claim to have experimentally established causation.

2. Scientific foundation

Conceptual integration research by Fauconnier and Turner examines how complex relations can be compressed into manageable conceptual structures [1]. Cowan's working-memory research examines the limited number of distinct chunks that can be maintained in attention, with learned chunks sometimes representing extensive information [2]. Chiang and Rich observed lower-dimensional, behaviorally biased neural representations during a spatial-memory chunking task in macaque prefrontal cortex, suggesting lossy compression in that specific task [3]. Owen and Manning reported relatively information-rich and compressible brain activity during more cognitively engaging conditions [4]. These findings support efficient biological representation but neither demonstrate complete hidden reasoning behind omitted verbal steps nor imply computational equivalence between brains and transformer language models.

Clark and Brennan's framework of conversational grounding establishes the importance of mutual knowledge to successful communication [5]. Inferential distance is used here as a descriptive term for the gap between stated concepts and what a listener needs to understand the intended relation, not as a standardized neurological metric.

Information-bottleneck theory formalizes the problem of retaining task-relevant information while compressing other information [6]. Applied as an analogy, prompt quality depends on whether essential constraints, dependencies, evidence, and authority boundaries survive compression, not simply on prompt length. This analogy does not describe the precise optimization procedure of either the human brain or a particular LLM.

In LLM research, Liu and colleagues demonstrated position-dependent weaknesses in the use of long-context evidence among tested models [7]. Laban and colleagues reported a mean performance decline of 39% across six evaluated generation tasks comparing multi-turn interaction with fully specified single-turn requests; premature assumptions and difficulty recovering from incorrect early outputs were prominent concerns [8]. The 39% figure is benchmark-specific, not an error rate for all LLM conversations. These studies support investigating the hypothesis, but do not prove compressed human thinking caused those effects.

3. Three different kinds of missing context

R - Recoverable omission: The prompt omits a relationship, but sufficient verified shared context is accessible. Success means reconstructing that supported relationship rather than inventing one.

U - Unrecoverable omission: A consequential fact is private, absent, or non-inferable. Success means making the uncertainty explicit, seeking clarification when warranted, or remaining within safe boundaries. A lucky guess does not qualify as reliable reconstruction.

P - Propagated mistaken inference: The model introduced an unsupported relationship in an earlier turn. Later turns test whether it recognizes the uncertainty and verifies or corrects the claim, or instead promotes it into an accepted premise.

Goal changes create a separate condition: during legitimate research, a human may approve a new objective after learning something. Such explicit revision is adaptation, not drift. A model silently changing requirements without authorization is drift. A valid experiment must distinguish these cases.

4. An intent-graph model

Let G_*t = (V_t, E_t, C_t, A_t) be the independently defined, currently approved task at conversation turn t. V represents goals and concepts; E their logical, causal, and dependency relationships; C constraints and evidence requirements; A authority and permitted-action boundaries. An approved change to the objective yields a new reference G*_t recorded in a change ledger. The model's interpretation, Ghat_t, is distinct from this reference.

The model's reconstruction can be described as Ghat_t = R(M_1:t, K_s, Ghat_(t-1)), where M_1:t denotes the conversation, K_s denotes accessible verified context, and R is the reconstruction process. This is an analytical account of behavior, not a description of model internals.

A proposed task-weighted deviation measure is:

$$D_t = (\Sigma_i w_i \times \delta_{i,t}) / \Sigma_i w_i$$

Here δ_i,t denotes disagreement with reference requirement i at turn t and w_i its predefined consequence weight. Critical safety and authority violations should weigh more than harmless formatting differences. The scoring plan must specify partial or uncertain answers and must avoid treating appropriate uncertainty about unrecoverable information as failure. The reference graph must not be generated solely from model outputs.

5. Context debt as an operational hypothesis

We define an assumption-promotion event as a model-generated claim not supported by available evidence or human confirmation that is later reused as a confirmed requirement, verified fact, or active premise without retaining its uncertainty.

The proposed context-debt measure is:

$$CD_t = \Sigma_a s_a \times \min(n_a, n_{max})$$

The sum covers active promoted but unverified assumptions. Each s_a denotes a predefined severity weight, n_a counts downstream decisions depending on assumption a, and n_max caps extreme dependency chains. When a claim is verified, rejected, or explicitly corrected, it no longer counts as

actively unverified, but its previous influence remains in the audit log.

This is a proposed measurement instrument, not a recognized psychometric or computational scale. Evaluators must test inter-rater agreement and whether the index predicts downstream task deviation after controlling for conversation length, task difficulty, and model differences. Deviation and context debt must be scored separately to avoid circular reasoning.

6. Illustrative industrial scenario

An engineer states: "The motor starts vibrating after product changeovers. Operators notice it before our sensors alarm. Build a predictive model, show it on a dashboard, and eventually expand it across the plant."

The explicit anchors are vibration, operator experience, prediction, visualization, and expansion. Missing dependencies include changeover conditions, sensor coverage, possible failure mechanisms, model target, acceptance criteria, and operational authority.

A model that diagnoses bearing failure and recommends automatic shutdown has introduced both an unsupported diagnosis and an unapproved action. It may produce an internally coherent answer while changing the actual project.

A possible safer reconstruction is: changeover conditions → attributed operator observations → sensor evidence → operating-state baseline → competing explanations → candidate predictive model → independent testing → read-only demonstration → human review → approved expansion.

This is a proposed hypothesis, not proof of the engineer's intent. Operator reports are attributable evidence, not automatic ground truth. Model conclusions must be separated from permitted actions. Should the engineer later approve a separate, governed control study, the reference objective can change through an explicit recorded decision. This scenario is illustrative, not a claim of a completed SynapticSteel experiment or a production control capability.

7. Five falsifiable hypotheses

H1 - Recoverability: Sparse prompts will produce more unsupported consequential inferences when required dependencies are unavailable than when shared evidence makes those omissions recoverable.

H2 - Propagation: Earlier incorrect inferences presented without uncertainty will be promoted and reused more often, and will predict greater subsequent deviation, than equivalent inferences clearly marked as unverified.

H3 - Context-fidelity intervention: Explicit labels such as USER-STATED, SOURCE-VERIFIED, MODEL-INFERRED and UNKNOWN, combined with targeted clarification, will reduce assumption promotion compared with a compressed baseline containing the same task facts.

H4 - Consequence sensitivity: Explicit tracking of authority boundaries will reduce critical violations more reliably than minor differences in wording or style.

H5 - Authorized revision and cost: The method will preserve approved goal changes while resisting unapproved changes, potentially at the cost of more clarifications, tokens, or time.

Null findings should be retained. An intervention may offer no advantage under matched information conditions; compressed prompts may work well when context is genuinely shared; and longer clarification exchanges are not necessarily beneficial.

8. Proposed controlled experiment: SynapticSteel Lab CC-001

Study status: NOT RUN. This is an experimental design, not a report of measurements.

Construct a non-sensitive task corpus containing independently specified intent graphs with verifiable outcomes and explicit prohibited actions. Include industrial condition monitoring, software architecture, research design, technical documentation, and general planning. For each task construct the three gap types R, U, and P, and include both unchanged goals and explicitly approved revisions.

Compare four prompt conditions:

A - Full specification. Supply complete relevant context as a reference upper bound.

B - Compressed baseline. Supply selected conceptual anchors with identified dependencies omitted.

C - Compressed plus reconstruction protocol. Supply exactly the same task facts as B, adding only instructions to label inferred relations, preserve constraints, and request clarification when consequential information is missing.

D - Compressed plus verified context anchor. Supply the sparse request and an independently verified goal-and-constraint record. D contains additional evidence and must be interpreted separately from the reasoning-only intervention in C.

Run both short and long fixed follow-up sequences, using isolated conversation starts, fixed model settings, randomized experimental condition order, and repeated trials. A twelve-task pilot could establish feasibility but should not be mistaken for a powered study. Pre-register the full sample size, inference settings, analysis plan, and scoring rubric before substantive evaluation.

Primary outcomes are D_t over turns, unsupported assumption rate, assumption promotion, and critical boundary violations. Secondary outcomes include CD_t, correct handling of unknowable context, recovery after correction, adaptation to approved revisions, task quality, and clarification costs. Reviewers should be blind to condition when feasible, and agreement and confidence intervals should be reported.

The strongest controlled comparison is B versus C, which keeps factual inputs equivalent. B versus D tests the effect of additional verified context, not the benefit of reconstruction instructions alone. The hypothesis is weakened if promoted assumptions do not predict later deviation, compression produces no excess consequential failures, or the intervention does not outperform a matched baseline.

No experiment, model training, paid API call, new service, or equipment action is authorized by this protocol.

9. Engineering interpretation

The immediate implication is a source-aware context-fidelity discipline, not a novel autonomous model subsystem. Consequential statements should be tagged as USER-STATED or APPROVED, SOURCE-VERIFIED, MODEL-INFERRED, UNKNOWN or CONTESTED, and SUPERSEDED WITH APPROVAL. Changes to the reference goal require clear human authorization. Summaries should preserve uncertainty and provenance rather than turning guesses into facts.

Two approaches are complementary. Human-side decompression makes critical dependencies explicit for high-stakes work. Model-side reconstruction proposes missing links while clearly identifying evidence gaps and seeking confirmation before promoting inferences into commitments. Neither approach authorizes independent industrial equipment control.

10. Limitations and conclusion

Human conceptual compression and LLM contextual reconstruction are distinct processes. The A-to-Z illustration is not a literal neural mechanism. The mathematical measures are proposed evaluation constructs, not validated scales. Model behavior may vary with task, architecture, context, and prompting. Human objectives can change appropriately over time, and clarification may impose friction.

The central question is narrow and falsifiable: Does unverified inference promotion, rather than compressed language alone, explain a measurable share of unapproved drift in human–LLM collaboration?

If so, improved reliability may require preserving boundaries between what the human stated, what the model inferred, what has been verified, and what has actually been approved. A reliable AI collaborator should not merely reach Z. It should preserve a traceable path to whichever Z the human authorizes.

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Publication note

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